

例题1:

石油在长度 $L=50\text{m}$ 、直径 $d=100\text{mm}$ 的管道内流动，管轴对于水平的倾斜度为 30° ， $\mu=0.285\text{Ns/m}^2$ ， $\rho=950\text{kg/m}^3$ ， $\text{Re}_{cr}=2000$ ，试确定

- (1) 为保持层流状态的最大允许流量；
- (2) 相应的进出口压差。

解: (1) 若流动管内层流 (即Hagen-Poiseuille流)

$$\text{由 } \text{Re}_{cr} = \frac{\rho \bar{V} d}{\mu} = 2000$$

$$\text{得到 } \bar{V} = 6 \text{ m/s}, Q = \bar{V} \frac{\pi d^2}{4} = 0.047 \text{ m}^3/\text{s}$$

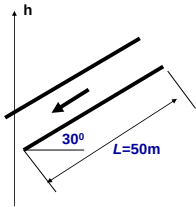
(2) 相应的进出口压差

$$\Delta p^* = \lambda \frac{1}{2} \rho \bar{V}^2 \frac{L}{d} = \frac{64}{\text{Re}_{cr}} \frac{1}{2} \rho \bar{V}^2 \frac{L}{d} = 273600 \text{ N/m}^2$$

$$\Delta p^* = \Delta p + \rho g \Delta h$$

$$\Delta h = L \sin \alpha$$

$$\Delta p = 273600 - \rho g \Delta h = 40850 \text{ N/m}^2$$

**例题2:**

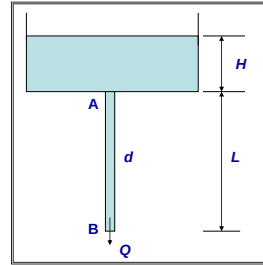
毛细管粘度计 (测 μ)

AB内近似看成定常充分发展层流

$$Q = -\frac{\pi d^4}{128 \mu} \frac{dp^*}{dx}$$

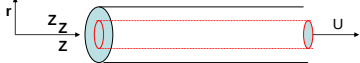
$$\begin{aligned} \frac{dp^*}{dx} &= -\frac{p_B + \rho g h_B - (p_A + \rho g h_A)}{L} \\ (\text{做了近似}) \quad &= \frac{p_A - p_B + \rho g (h_A - h_B)}{L} \\ &= \frac{\rho g H + \rho g L}{L} \end{aligned}$$

$$\therefore \mu = \frac{\pi d^4}{128 Q} \frac{\rho g (H + L)}{L}$$



近似: 无限长, 至少 $L/d > 100$, 入口效应可不计了
定常, H 应恒定, 实际选取一段时间内平均高度近似
层流, d 是毫米量级, 流速很慢, $\text{Re} < 2000$
圆管, 细选毛细管, 横截面一定
理论公式, 如何在实际中应用

例题3: 无限长的半径为 R_1 的圆柱, 以常速度 U 沿轴线方向在无限长的半径为 R_2 的同轴圆柱面内运动, 试确定由此引起的两柱面间的粘性不可压缩流体运动的速度剖面 (质量力略去不计, 且已知轴向压力梯度为零)



解: 采用柱坐标系 (r, θ, z) 轴与轴线重合

由题意知: $V_r = V_\theta = 0$ $V_z = V$

$$\frac{\partial}{\partial t} = 0 \quad \frac{\partial}{\partial \theta} = 0 \quad \frac{\partial}{\partial z} = 0$$

由连续方程知: $\frac{\partial V}{\partial r} = 0 \Rightarrow V = V(r)$

由径向运动方程知: $\frac{\partial p}{\partial r} = 0$ 周向运动方程自动满足

由轴向运动方程知: $\frac{d}{dr} \left(r \frac{dV}{dr} \right) = 0$ $c_1 = -\frac{U}{\ln(R_2/R_1)}$

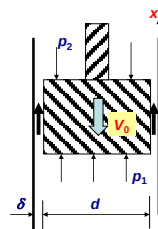
积分两次得: $V = c_1 \ln r + c_2$

边界条件: $V|_{r=R_1} = U$ $V|_{r=R_2} = 0$

所以: $V = U \frac{\ln(r/R_2)}{\ln(R_1/R_2)}$

例题4: 充满油的油缸内有长 L 活塞向下以 V_0 等速运动, 油通过间隙 δ 向上流动, $\delta \ll d$, 流动是层流, 已知油粘度系数 μ , 密度 ρ , 活塞上下压差 $p_2 - p_1$, 活塞直径 d , 间隙 δ , 忽略端部效应, 求 V_0 。

解: 环形通道, $\delta \ll d$, 缝隙内看成无限大两平板间的Couette流动



缝隙内流动方向为 x 向, $U = -V_0$ (要求的量)

$$Q = V_0 \frac{\pi}{4} d^2$$

$$[\text{couette } Q = b \int_0^H u dy = b (-\frac{1}{12\mu} \frac{dp^*}{dx} H^3 + \frac{UH}{2})]$$

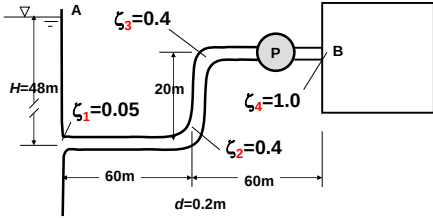
$$\delta \ll d \quad b = \pi d, \quad H = \delta$$

$$Q = \pi d (-\frac{1}{12\mu} \frac{dp^*}{dx} \delta^3 - \frac{V_0 \delta}{2}) = V_0 \frac{\pi}{4} d^2 \quad \text{①}$$

$$\text{其中 } \frac{dp^*}{dx} = \frac{\Delta(p + \rho g h)}{L} = \frac{(p_2 + \rho g L) - p_1}{L} = \frac{p_2 - p_1}{L} + \rho g$$

$$\text{由①得到 } V_0 \left(\frac{\delta^3}{2} + \frac{\pi d^2}{4} \right) = -\frac{\delta^3}{12\mu} (p_2 - p_1 + \rho g L)$$

$$V_0 = \frac{\delta^3}{3\mu(2\delta + \pi d^2)} \frac{(p_1 - p_2 - \rho g L)}{L} \approx \frac{\delta^3}{3\mu\pi d^2 L} (p_1 - p_2 - \rho g L)$$



例题7

已知: $\mu=1.14 \times 10^{-3} \text{m}^2/\text{s}$, $\rho=1000 \text{kg}/\text{m}^3$, 粗糙度 $h/d=0.00023$, $N_{\text{pump}}=20 \text{kW}$, $Q=0.14 \text{m}^3/\text{s}$
求: B处的压力(表压)

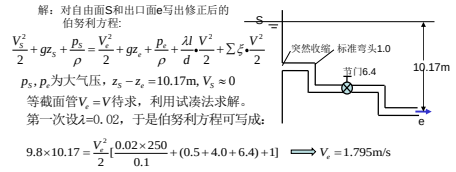
$$p_A + \frac{1}{2} \rho V_A^2 + \rho g h_A + \Delta p_{\text{pump}} = p_B + \frac{1}{2} \rho V_B^2 + \rho g h_B + \Delta p_f + \sum \Delta p_\zeta$$

$$\Delta p_{\text{pump}} = \frac{N_{\text{pump}}}{Q} \quad V = V_B = \frac{Q}{\frac{\pi}{4} d^2} \quad Re = \frac{\rho V d}{\mu} = 7.81 \times 10^5 \quad \lambda = 0.0152$$

$$\Delta p_f = \sum_{i=1}^3 \lambda \frac{L_i}{d_i} \frac{\rho V_i^2}{2} \quad \sum \Delta p_\zeta = (\zeta_1 + \zeta_2 + \zeta_3 + \zeta_4) \frac{\rho V^2}{2} \quad p_B - p_a = 283 \text{kPa}$$

(三) 管道系统流动计算

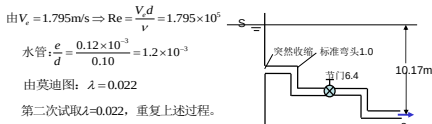
2. 试确定如图所示的管路系统的流量Q, 已知水温 $t=25^\circ\text{C}$,
水管内径为 $d=10 \text{cm}$, 长度为 $l=250 \text{m}$, 绝对粗糙度 $e=0.12 \text{mm}$.



$$9.8 \times 10.17 = \frac{V^2}{2} \left(\frac{0.02 \times 250}{0.1} + (0.5 + 4.0 + 6.4) + 1 \right) \Rightarrow V = 1.795 \text{m/s}$$

6

(三) 管道系统流动计算

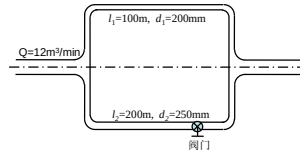


由 $V_e = 1.795 \text{m/s} \Rightarrow Re = \frac{V_e d}{\nu} = 1.795 \times 10^5$
水管: $\frac{e}{d} = \frac{0.12 \times 10^{-3}}{0.10} = 1.2 \times 10^{-3}$
由莫迪图: $\lambda = 0.022$
第二次试取 $\lambda = 0.022$, 重复上述过程。
得: $V_e = 1.726 \text{m/s}$, $Re = \frac{1.726 \times 0.1}{10^{-6}} = 1.726 \times 10^5$
由莫迪图查出 $\lambda = 0.022$, 因此经过两次试凑获得足够的精度。
通过的流量: $Q = VA = \frac{\pi}{4} \times 0.1^2 \times 1.726 = 0.01356 \text{m}^3/\text{s} = 13.56 \text{ l/s}$

7

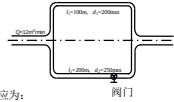
(三) 管道系统流动计算

3. 水平安装的铸铁管网(绝对粗糙度 $e=0.26 \text{mm}$) 如图所示,
第一分支长度 $l_1=100 \text{m}$, 管径 $d_1=200 \text{mm}$,
第二分支长度 $l_2=200 \text{m}$, 管径 $d_2=250 \text{mm}$, 在第二分支上有半开的阀门。
已知总流量为 Q , $Q=12 \text{m}^3/\text{min}$, 试确定两分支上的流量及分支两端的压差。
水温为 20°C , 不计岔管和弯头的局部损失, 阀门的局部损失系数为 4.5 。



8

(三) 管道系统流动计算



解：对于任一分支而言（ $i=1,2$ ），压降均应为：

$$\Delta p_i = \lambda_i \frac{l_i}{d_i} \frac{\rho V_i^2}{2} + \sum \xi_i \frac{\rho V_i^2}{2} \quad \text{式中：} \quad V_i = \frac{Q_i}{\pi d_i^2/4}$$

$$\text{代入得：} \quad \Delta p_i = \frac{8\rho}{\pi^2} \left(\lambda_i \frac{l_i}{d_i} + \sum \xi_i \right) \frac{Q_i^3}{d_i^5}$$

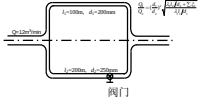
$$\text{由 } \Delta p_1 = \Delta p_2, \quad \sum \xi_i = 0 \Rightarrow \frac{Q_2}{Q_1} = \left(\frac{d_1}{d_2} \right)^2 \sqrt{\frac{\lambda_2 l_2/d_2 + \sum \xi_2}{\lambda_1 l_1/d_1}}$$

采用试凑法求解，取 $\lambda_1 = \lambda_2 = 0.02$ ，因此

$$\lambda_2 l_2/d_2 + \sum \xi_2 = 0.02 \times \frac{200}{0.25} + 4.5 = 20.5$$

$$\lambda_1 l_1/d_1 = 0.02 \times \frac{100}{0.2} = 10$$

(三) 管道系统流动计算



$$\text{于是得：} \quad \frac{Q_2}{Q_1} = \left(\frac{0.2}{0.25} \right)^2 \sqrt{\frac{20.5}{10}} = 0.916$$

$$\text{而 } Q = Q_1 + Q_2, \quad \text{即 } Q/Q_1 = Q_1/Q_2 + 1 = 1.916$$

$$Q_2 = \frac{12/60}{1.916} = 0.1044 \text{ m}^3/\text{s}, \quad Q_1 = 0.2 - 0.1044 = 0.0956 \text{ m}^3/\text{s}$$

$$\text{于是：} \quad V_2 = \frac{0.1044}{(\pi/4)0.25^2} = 2.13 \text{ m/s}$$

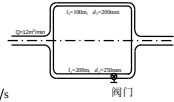
$$\text{由于 } \nu = 1.00 \times 10^{-6} \text{ m}^2/\text{s} \text{ (在 } t=20^\circ\text{C} \text{ 条件下)} \quad \text{Re}_2 = \frac{V_2 d_2}{\nu} = \frac{2.13 \times 0.25}{1.00 \times 10^{-6}} = 5.32 \times 10^5$$

$$\text{铸铁管的 } e=0.26 \text{ mm, 于是：} \quad \frac{e}{d_2} = \frac{0.00026}{0.25} = 0.001$$

由莫迪图： $\lambda_2 = 0.02$ ，此值正是试凑所取之值，于是

$$\Delta p_2 = \left(\lambda_2 \frac{l_2}{d_2} + \xi_2 \right) \frac{\rho V_2^2}{2} = 20.5 \times \frac{998.3 \times 2.13^2}{2} = 46.5 \text{ kPa}$$

(三) 管道系统流动计算



$$\text{同理：} \quad V_1 = \frac{Q_1}{\pi d_1^2/4} = \frac{0.0956}{(\pi/4)0.1^2} = 3.04 \text{ m/s}$$

$$\text{Re}_1 = \frac{V_1 d_1}{\nu} = \frac{3.04 \times 0.1}{1.00 \times 10^{-6}} = 6.09 \times 10^5 \quad \frac{e}{d_1} = \frac{0.00026}{0.1} = 0.0013$$

由莫迪图： $\lambda_1 = 0.0208$

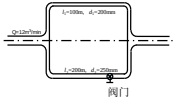
$$\Delta p_1 = \lambda_1 \frac{l_1}{d_1} \frac{\rho V_1^2}{2} = 10 \times \frac{998.3 \times 3.04^2}{2} = 48.0 \text{ kPa}$$

此值与 Δp_2 相比，尚应修正试凑值。

可取 $Q_2 = 0.105 \text{ m}^3/\text{s}$ ， $Q_1 = 0.095 \text{ m}^3/\text{s}$ ，于是可得：

$$V_2 = \frac{Q_2}{\pi d_2^2/4} = \frac{0.105}{(\pi/4)0.25^2} = 2.14 \text{ m/s}$$

(三) 管道系统流动计算



$$\text{Re}_2 = \frac{V_2 d_2}{\nu} = \frac{2.14 \times 0.25}{1.00 \times 10^{-6}} = 5.34 \times 10^5$$

由莫迪图： $\lambda_2 = 0.02$

$$\Delta p_2 = \left(\lambda_2 \frac{l_2}{d_2} + \xi_2 \right) \frac{\rho V_2^2}{2} = 20.5 \times \frac{998.3 \times 2.14^2}{2} = 47 \text{ kPa}$$

$$\text{同理：} \quad V_1 = \frac{Q_1}{\pi d_1^2/4} = \frac{0.095}{(\pi/4)0.1^2} = 3.02 \text{ m/s}$$

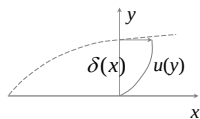
$$\text{Re}_1 = 6.05 \times 10^5, \quad \lambda_1 = 0.0208, \quad \Delta p_1 = 47.3 \text{ kPa}$$

故真实解为： $Q_2 = 0.105 \text{ m}^3/\text{s}$ ， $Q_1 = 0.095 \text{ m}^3/\text{s}$

分支两端压差为： $\Delta p = 47 \text{ kPa}$ 。

例题

$$\frac{u}{U} = a + b \frac{y}{\delta} + c \left(\frac{y}{\delta} \right)^2$$



$$\begin{aligned} BC: \quad y=0 \quad u=0 \\ y=\delta \quad u \approx U \quad \frac{\partial u}{\partial y} \approx 0 \quad a=0, b=\frac{2U}{\delta}, c=-\frac{U}{\delta^2} \end{aligned}$$

$$\frac{u}{U} = \frac{2y}{\delta} - \frac{y^2}{\delta^2} \quad 0 \leq y \leq \delta(x)$$

$$\delta_2 = \int_0^\delta \frac{u}{U} \left(1 - \frac{u}{U}\right) dy \quad \frac{u}{U} = \frac{2y}{\delta} - \frac{y^2}{\delta^2}$$

$$\delta_2 = \int_0^\delta \left(\frac{2y}{\delta} - \frac{y^2}{\delta^2} \right) \left(1 - \frac{2y}{\delta} + \frac{y^2}{\delta^2} \right) dy \quad \text{let } \eta = \frac{y}{\delta}$$

$$\delta_2 = \delta \int_0^1 (2\eta - \eta^2)(1 - 2\eta + \eta^2) d\eta$$

$$= \delta \int_0^1 (-\eta^4 + 4\eta^3 - 5\eta^2 + 2\eta) d\eta = \frac{2}{15} \delta \quad \delta_2 = \frac{2}{15} \delta$$

$$\tau_w = \mu \frac{\partial u}{\partial y} \Big|_{y=0} = \frac{\mu U}{\delta} \frac{\partial (u/U)}{\partial (y/\delta)} \Big|_{y=0} \quad \tau_w(x) = \frac{2\mu U}{\delta}$$

例：有一块 $1.5 \text{ m} \times 4.5 \text{ m}$ 的矩形薄板在空气中以 3 m/s 速度沿板面方向拖动，已知空气运动粘性系数为 $\nu = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$ ，密度为 $\rho = 1.2 \text{ kg/m}^3$ 。试求薄板沿短边方向和长边方向运动时，各自的摩擦阻力。

解：（1）求沿短边方向运动的阻力。

$$\text{Re}_L = \frac{UL}{\nu} = \frac{3 \times 1.5}{1.5 \times 10^{-5}} = 3 \times 10^5$$

$$\text{转捩点: } x_k = \text{Re}_{x_k} \frac{\nu}{U_0} = 5 \times 10^5 \times \frac{1.5 \times 10^{-5}}{3} = 2.5 \text{ m}$$

因为 $L_1 = 1.5 \text{ m} < 2.5 \text{ m}$ ，所以全板为层流边界层，

$$C_{f_1} = \frac{1.33}{\text{Re}_{L_1}^{0.5}} = \frac{1.33}{\sqrt{\frac{3 \times 1.5}{1.5 \times 10^{-5}}}} = 0.00242$$

$$D_{f_1} = 2C_{f_1} b L \frac{\rho}{2} U_0^2 = 2 \times 0.00242 \times 4.5 \times 1.5 \times \frac{1.2}{2} 3^2 = 0.176 \text{ N}$$

（2）求沿长边运动时的阻力

$L_2 > x_k$ ，为混合附面层

$$\text{Re}_{L_2} = \frac{U_0 L_2}{\nu} = \frac{3 \times 4.5}{1.5 \times 10^{-5}} = 9 \times 10^5 < 10^6$$

$$\begin{aligned} C_{f_2} &= \frac{0.074}{\text{Re}_{L_2}^{0.2}} - \frac{A}{\text{Re}_{L_2}} \\ &= \frac{0.074}{(9 \times 10^5)^{0.2}} - \frac{1700}{9 \times 10^5} = 0.0288 \end{aligned}$$

$$D_{f_2} = 2C_{f_2} b L_2 \frac{\rho}{2} U_0^2 = 2 \times 0.0288 \times 1.5 \times 4.5 \times \frac{1.2}{2} 3^2 = 0.21 \text{ N}$$

比较（1），（2）结果，一块平板混合边界层产生的阻力大于层流边界层的阻力。